PHY101(Quiz#04)
27 Dec 2024

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 08:03 AM, 27 December 2024**

Question # 1 of 10 (Start time: 08:03:54 AM, 27 December 2024) **Total Marks:** 1

In the given formula, $\Delta U = \gamma \times \Delta A$, what does each variable represent?

Select the correct option

Reload Math Equations

- ○ ΔU = Energy per unit area, γ = Force, ΔA = Area
- ◉ ΔU = Change in potential energy, γ = Surface tension, ΔA = Change in area
- ○ ΔU = Work done, γ = Length, ΔA = Acceleration
- ○ ΔU = Velocity, γ = Length, ΔA = Change in area

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 08:03 AM, 27 December 2024

Question # 2 of 10 (Start time: 08:05:19 AM, 27 December 2024) **Total Marks:** 1

Energy density in case of a capacitor is always proportional to:

Select the correct option

Reload Math Equations

- ○ E^2
- ○ CV
- ○ C^2
- ◉ V^2

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 08:03 AM, 27 December 2024

Question # 3 of 10 (Start time: 08:06:24 AM, 27 December 2024) **Total Marks:** 1

The S.I unit of stress is same as that of

Select the correct option

Reload Math Equations

- ○ Impulse
- ◉ Force per unit area
- ○ Force
- ○ Momentum

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 08:03 AM, 27 December 2024

Question # 4 of 10 (Start time: 08:07:15 AM, 27 December 2024) **Total Marks:** 1

Two skaters toss a basketball back and forth on frictionless ice. Which of the following does not change?

Select the correct option

○	the momentum of the basketball;
●	the momentum of the system consisting of both skaters and the basketball
○	the momentum of the system consisting of one skater and the basketball;
○	the momentum of an individual skater;

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 08:03 AM, 27 December 2024**

Question # 5 of 10 (Start time: 08:08:15 AM, 27 December 2024) **Total Marks:** 1

In .....object returns to its original position if displaced slightly

Select the correct option

- ○ unstable equilibrium
- ◉ stable equilibrium
- ○ rotational equilibrium
- ○ dynamic equilibrium

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 08:03 AM, 27 December 2024

Question # 6 of 10 (Start time: 08:09:10 AM, 27 December 2024) **Total Marks:** 1

A force of 10 N is applied to push a box 4 m across a floor. What is the work done on the box if the force is applied in the same direction as the motion?

Select the correct option

Reload Math Equations

- ○ 50 J
- ◉ 40 J
- ○ 30 J
- ○ 20 J

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 08:03 AM, 27 December 2024**

Question # 7 of 10 (Start time: 08:09:42 AM, 27 December 2024) **Total Marks:** 1

How is the rotational inertia about an axis perpendicular to the xy-plane calculated?

Where, I= moment of inertia, m= mass, r=distance

Select the correct option

Reload Math Equations

- ○ $I = mr^2/2$
- ○ $I = m/r$
- ◉ $I = mr^2$
- ○ $I = mr$

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 08:10:17 AM, 27 December 2024) **Total Marks:** 1

If the frequency of a sound wave increases due to the Doppler effect, what happens to the wavelength?

Select the correct option

Reload Math Equations

- ○ It increases.
- ◉ It decreases.
- ○ It becomes undefined.
- ○ It remains constant.

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 08:03 AM, 27 December 2024**

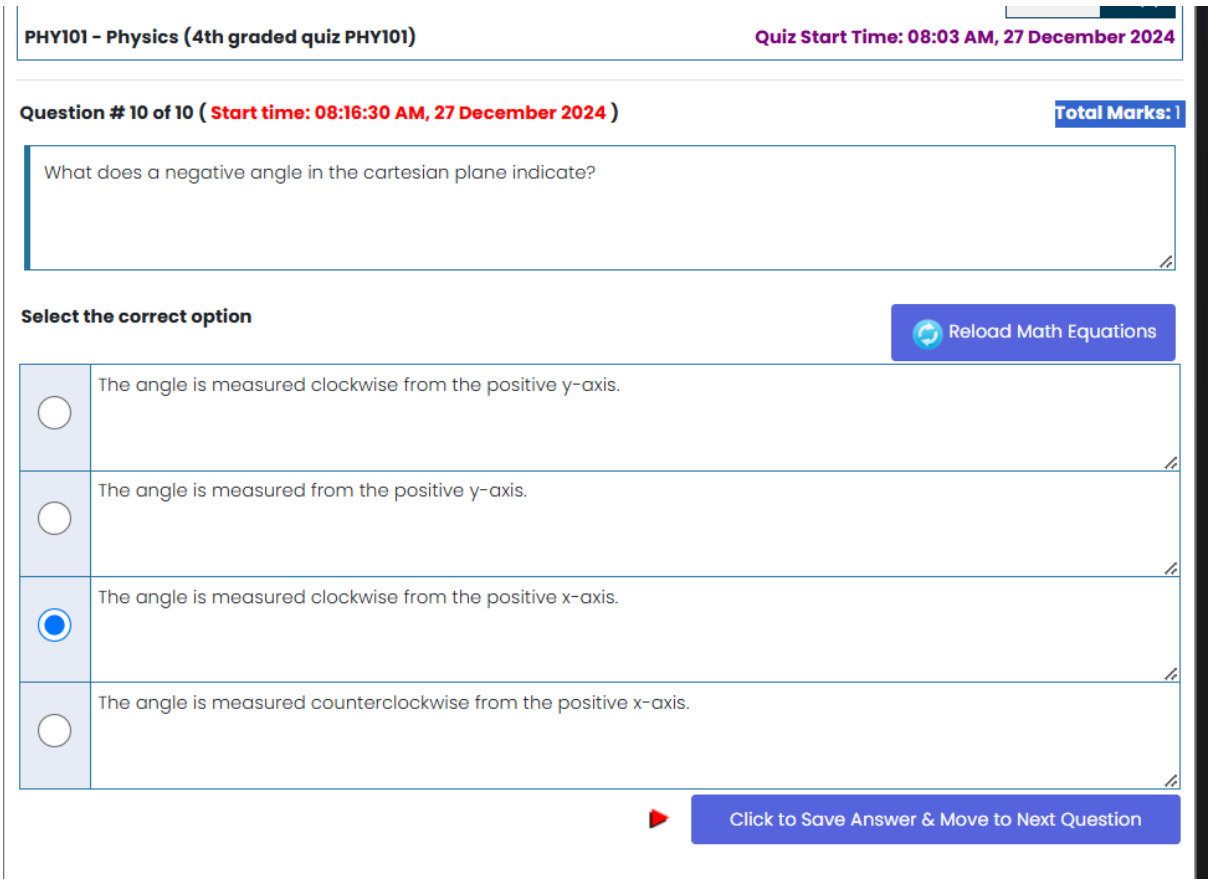
Question # 9 of 10 (Start time: 08:15:36 AM, 27 December 2024) **Total Marks:** 1

A student rubs a rod held in his hand.
Which action causes the rod to gain a large electrostatic charge?

Select the correct option

○	Rubbing an iron rod with a woolen duster
◉	Rubbing a polythene rod with a woolen duster
○	Rubbing a polythene rod with a steel magnet
○	Rubbing an iron rod with a steel magnet

Click to Save Answer & Move to Next Question



FROM
Ma’am Inaayat

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PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 01:45 PM, 27 December 2024

Question # 1 of 10 (Start time: 01:46:00 PM, 27 December 2024) **Total Marks:** 1

Frequency is defined as:

Select the correct option

Reload Math Equations

- ○ The energy of the oscillating object
- ○ The time to complete one cycle
- ○ The force needed to sustain oscillation
- ◉ The number of cycles per second

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 01:45 PM, 27 December 2024

Question # 2 of 10 (Start time: 01:47:03 PM, 27 December 2024) **Total Marks:** 1

Which of the following will remain unchanged when a sound wave travels in air or in water?

Select the correct option

- ○ Amplitude
- ◉ Frequency
- ○ Wavelength
- ○ Speed

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 01:45 PM, 27 December 2024**

Question # 3 of 10 (Start time: 01:47:33 PM, 27 December 2024) **Total Marks:** 1

In the context of magnetic force, if the velocity vector v is perpendicular to the magnetic field vector B, the resulting force vector F will be:

Select the correct option

Reload Math Equations

○	Can not be determined
●	Maximum
○	Zero
○	Minimum

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 01:45 PM, 27 December 2024

Question # 4 of 10 (Start time: 01:48:00 PM, 27 December 2024) **Total Marks:** 1

When a particle moving along a circular path, its projection along the diameter executes:

Select the correct option

Reload Math Equations

◉	Simple hormonic motion
○	Vibratory motion
○	Rotatory motion
○	Linear motion

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 01:45 PM, 27 December 2024

Question # 5 of 10 (Start time: 01:48:31 PM, 27 December 2024) **Total Marks:** 1

In a non-rigid rotating object, particles may have:

Select the correct option

Reload Math Equations

○	Zero momentum
○	No angular momentum
◉	Different rotational speeds
○	Identical angular velocities

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 01:49:01 PM, 27 December 2024) **Total Marks:** 1

Lenz's law is in accordance with law of conservation of

Select the correct option

○	Momentum
◉	Energy
○	Mass
○	Charge

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 01:45 PM, 27 December 2024

Question # 7 of 10 (Start time: 01:49:52 PM, 27 December 2024) **Total Marks:** 1

If a system is undergoing forced oscillation and has a high damping coefficient, how does this affect resonance?

Select the correct option

Reload Math Equations

○	Changes the phase angle of resonance
○	No effect on resonance
○	Increases resonance
◉	Decreases resonance

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 01:45 PM, 27 December 2024

Question # 8 of 10 (Start time: 01:50:23 PM, 27 December 2024) **Total Marks:** 1

Which of the following is an example of a scalar quantity?

Select the correct option

Reload Math Equations

- ○ Force
- ○ Displacement
- ○ Velocity
- ◉ Temperature

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 01:45 PM, 27 December 2024

Question # 9 of 10 (Start time: 01:50:52 PM, 27 December 2024) **Total Marks:** 1

Newton's second law of motion is also called:

Select the correct option

- ○ Law of gravitation
- ○ Law of torque
- ○ Law of inertia
- ◉ Law of acceleration

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 01:45 PM, 27 December 2024

Question # 10 of 10 (Start time: 01:51:25 PM, 27 December 2024) **Total Marks:**

A charge of 30 C flows through an electric appliance in 2.0 minutes. What is the average current in the appliance?

Select the correct option

- 4.0 A
- 15 A
- 0.25 A
- 60 A

Click to Save Answer & Move to Next Question

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PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 07:20 AM, 28 December 2024

Question # 1 of 10 (Start time: 07:20:04 AM, 28 December 2024) **Total Marks:** 1

A transformer is employed to:

Select the correct option

Reload Math Equations

○	Convert D.C. into A.C.
○	Obtain a suitable D.C. voltage.
◉	Obtain a suitable A.C. voltage.
○	Convert A.C. into D.C.

Click to Save Answer & Move to Next Question

sec(s)

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 07:20 AM, 28 December 2024**

Question # 2 of 10 (Start time: 07:21:01 AM, 28 December 2024) **Total Marks:** 1

In the scalar product of vectors, A and B, what is the resultant product?

Select the correct option

Reload Math Equations

- ○ A matrix
- ○ A vector
- ◉ A scalar
- ○ A quaternion

Click to Save Answer & Move to Next Question

sec(s)

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 07:20 AM, 28 December 2024**

Question # 3 of 10 (Start time: 07:21:32 AM, 28 December 2024) **Total Marks:** 1

For a mass-spring system, what is the relationship between the force F(x) = -kx and the stored energy $U(x) = \frac{1}{2}kx^2$?

Select the correct option

Reload Math Equations

- ○ The force and energy are inversely proportional
- ◉ The relationship is non-linear
- ○ The force and energy are directly proportional
- ○ There is no relationship between force and energy

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 07:20 AM, 28 December 2024

Question # 4 of 10 (Start time: 07:22:22 AM, 28 December 2024) **Total Marks:** 1

What is the primary reason a rocket moves forward when it expels exhaust gases?

Select the correct option

Reload Math Equations

- (●) Reaction force according to Newton's Third Law
- ( ) Decrease in air resistance
- ( ) Gravitational pull
- ( ) Increase in mass

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 07:20 AM, 28 December 2024

Question # 5 of 10 (Start time: 07:22:53 AM, 28 December 2024) **Total Marks:** 1

If you double the height of an object above the Earth's surface, how does the gravitational potential energy change?

Select the correct option

Reload Math Equations

- (●) It doubles
- ( ) It halves
- ( ) It remains constant
- ( ) It becomes zero

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 07:20 AM, 28 December 2024

Question # 6 of 10 (Start time: 07:23:24 AM, 28 December 2024) **Total Marks:** 1

What mathematical tool represents the rate of change of a function?

Select the correct option

Reload Math Equations

- ○ Logarithm
- ○ Exponent
- ◉ Derivative
- ○ Integral

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 07:20 AM, 28 December 2024

Question # 7 of 10 (Start time: 07:23:56 AM, 28 December 2024) **Total Marks:** 1

We will apply the condition of equilibrium to situations in which all forces are:

Select the correct option

Reload Math Equations

○	Upward
○	Downward
◉	Coplanar
○	Straight line

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 07:20 AM, 28 December 2024**

Question # 8 of 10 (Start time: 07:24:27 AM, 28 December 2024) **Total Marks:** 1

What does the modern view refer to as "inertia"?

Select the correct option

Reload Math Equations

- (●) The resistance to changes in motion
- ( ) The tendency of objects to accelerate
- ( ) The force required to move an object
- ( ) The tendency of objects to decelerate

Click to Save Answer & Move to Next Question

PHY101 - Physics (4th graded quiz PHY101) **Quiz Start Time: 07:20 AM, 28 December 2024**

Question # 9 of 10 (Start time: 07:24:58 AM, 28 December 2024) **Total Marks:** 1

Magnetic flux (Φ_B) is defined as:

Select the correct option

Reload Math Equations

- (●) The quantity of magnetic field passing through a surface
- ( ) The ability of a material to conduct magnetic lines
- ( ) The measure of magnetic pole strength
- ( ) The strength of a magnet

Click to Save Answer & Move to Next Question

sec(s)

PHY101 - Physics (4th graded quiz PHY101) Quiz Start Time: 07:20 AM, 28 December 2024

Question # 10 of 10 (Start time: 07:25:32 AM, 28 December 2024) **Total Marks:** 1

How does the period (T) change if the the linear speed (v) is doubled while keeping radius (R) of the circular path constant?

Select the correct option

Reload Math Equations

- ○ Remains the same
- ○ Increases
- ○ Cannot be determined
- ◉ Decreases

Click to Save Answer & Move to Next Question

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